

Employee Attrition Analysis using Data Mining Techniques

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INTRODUCTION

Every year a lot of companies hire a number of employees. Acquiring a new employee can become costly in both time and money, therefore some companies are going great lengths trying to keep their employees happy and satisfied. One of the ways how to keep employees from leaving is to analyze, why people left the company and predict who could be leaving next and try to take preventive action. We are aiming to develop a predictive model using suitable classifier for 'Attrition', which can also provide importance of features in the model.

DATA DESCRIPTION

The given data is regarding 4400 employees in a company, whether they have left the company in the last year (Attrition : Yes or No). The basic information of the employees includes variables like age, education and income. The responses in manager and employee surveys are included which take into account the factors like job satisfaction, performance and work - life balance.

The in time and out times recorded daily over the year 2015 (01-01-2015 to 31-12-2015) are available for each employee. The following two variables are created using both these times and used for further analysis.

1) Number of Working Days :

The days in the year 2015 for which In time (or Out time) is positive i.e. not NA. These reflect the variation in the working days excluding the common holidays like weekends and festivals.

Min	Q1	Q2	Q3	Max	Mean
225.0	232.0	236.0	241.0	248.0	236.3

2) Average Working Hours :

The difference between corresponding Out time and In time gives the working time (in hours) for the day. Average over such working times for all working days in the year is considered.

Min	Q1	Q2	Q3	Max	Mean
5.951	6.673	7.407	8.368	11.031	7.701

The range for the average working hours is observed to be very high.

EXPLORATORY ANALYSIS

Does Travelling affect Attrition?

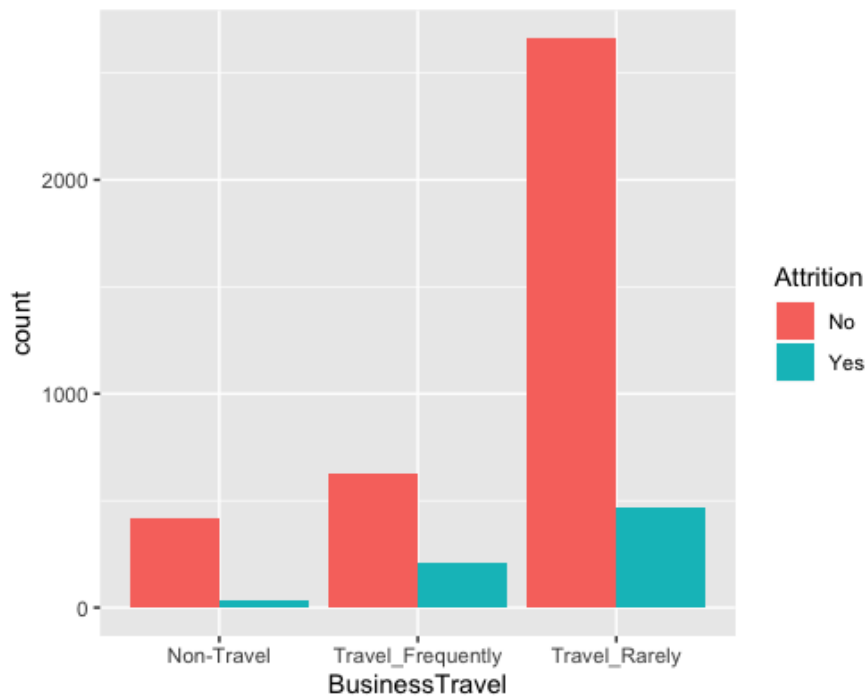


Figure 1

There is not much difference between the attrition of employees with different travelling behaviour. Majority of the employees travel rarely.

Which Department has highest Attrition?

Attrition by Department

About 30% employees from the HR department leave the job or get fired. This proportion is almost double than the other two departments. This is not surprising as generally there are lot of opportunities in the market for HR people than the other two.

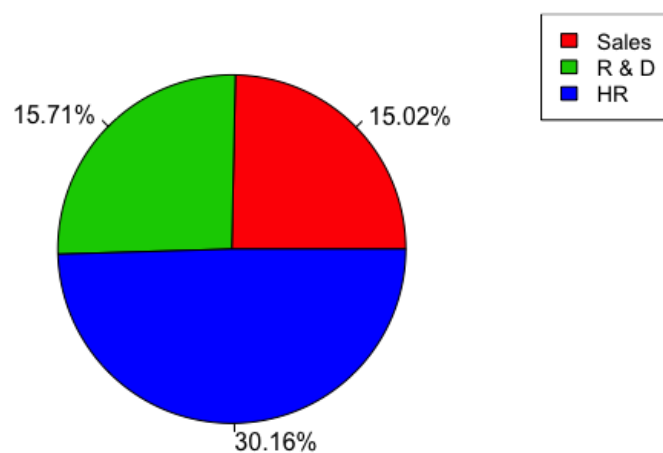


Figure 2

Relation with Working Hours, Age, Working Years and Monthly Income

Attrition	Avg. Working Hrs	Age	Working Years	Monthly Income
Yes	8.32	33.61	8.26	61682
No	7.58	37.56	11.86	65672
p-value	0	0	0	0.03

Table 1

Means of each variable are calculated for both the class of employees. Also, two sample t-test is performed on each variable to check whether means differ significantly for different Attrition (Yes or No). From the p-values, we can see that means of Working Hrs, Age and Working Years are significantly different for different attrition level. While, Monthly income is also different, but significance is not as strong as of previous variables. All these can also be seen from the following Box plots.

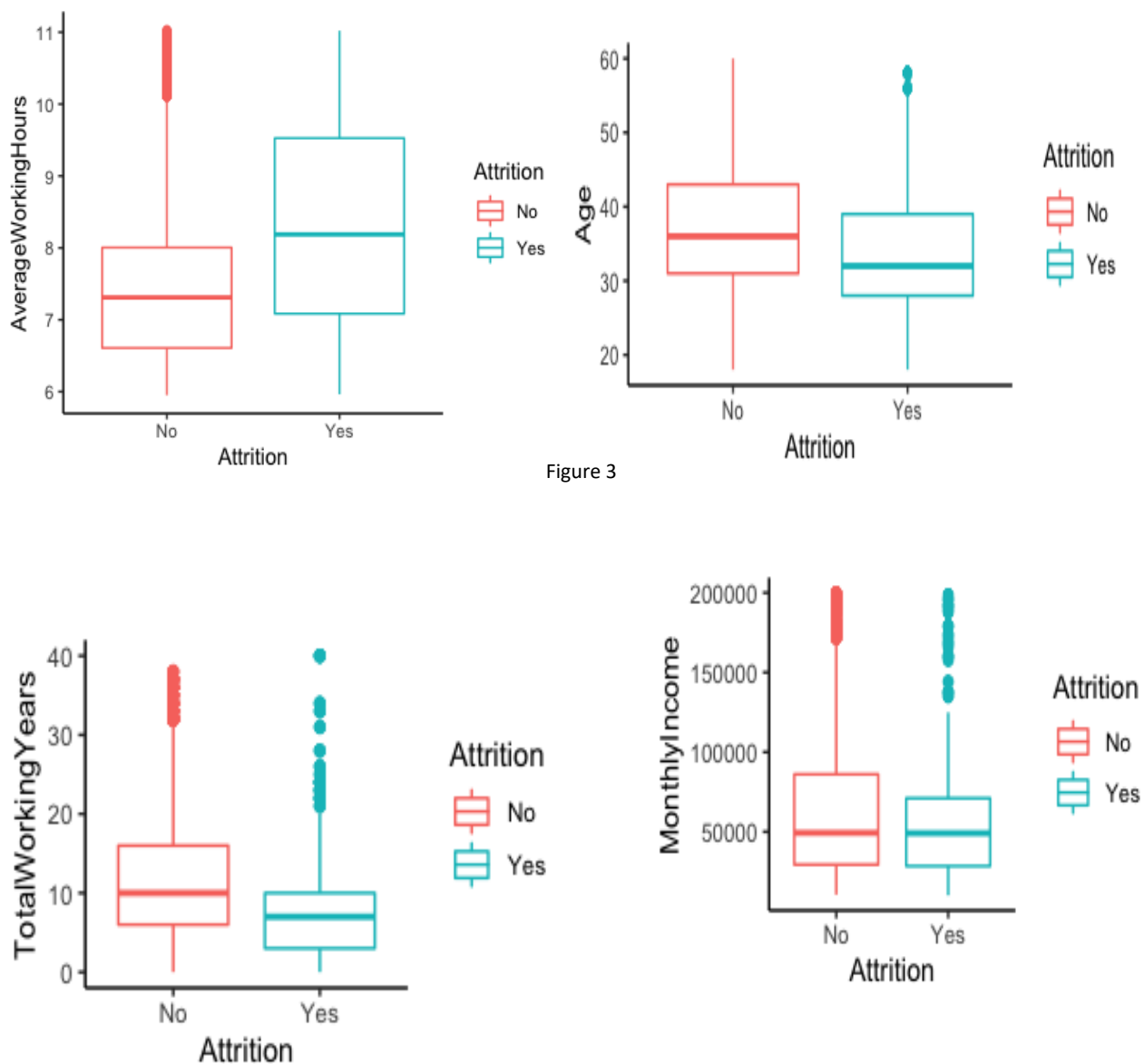


Figure 4

Building a Predictive Model

The Random forest works on Bagging principle. It is an ensemble of Decision Trees. The bagging method is used to increase the overall results by combining weak models. The model considers 6 randomly selected variables each time. The parameter number of variables (mtry) value was decided using tuning. Random forest also doesn't over fit easily because of its randomness feature. Random Forest helps in identification of important variables in classification by ranking them according to the mean decrease in Gini index. The Variable Importance plot is as follows:

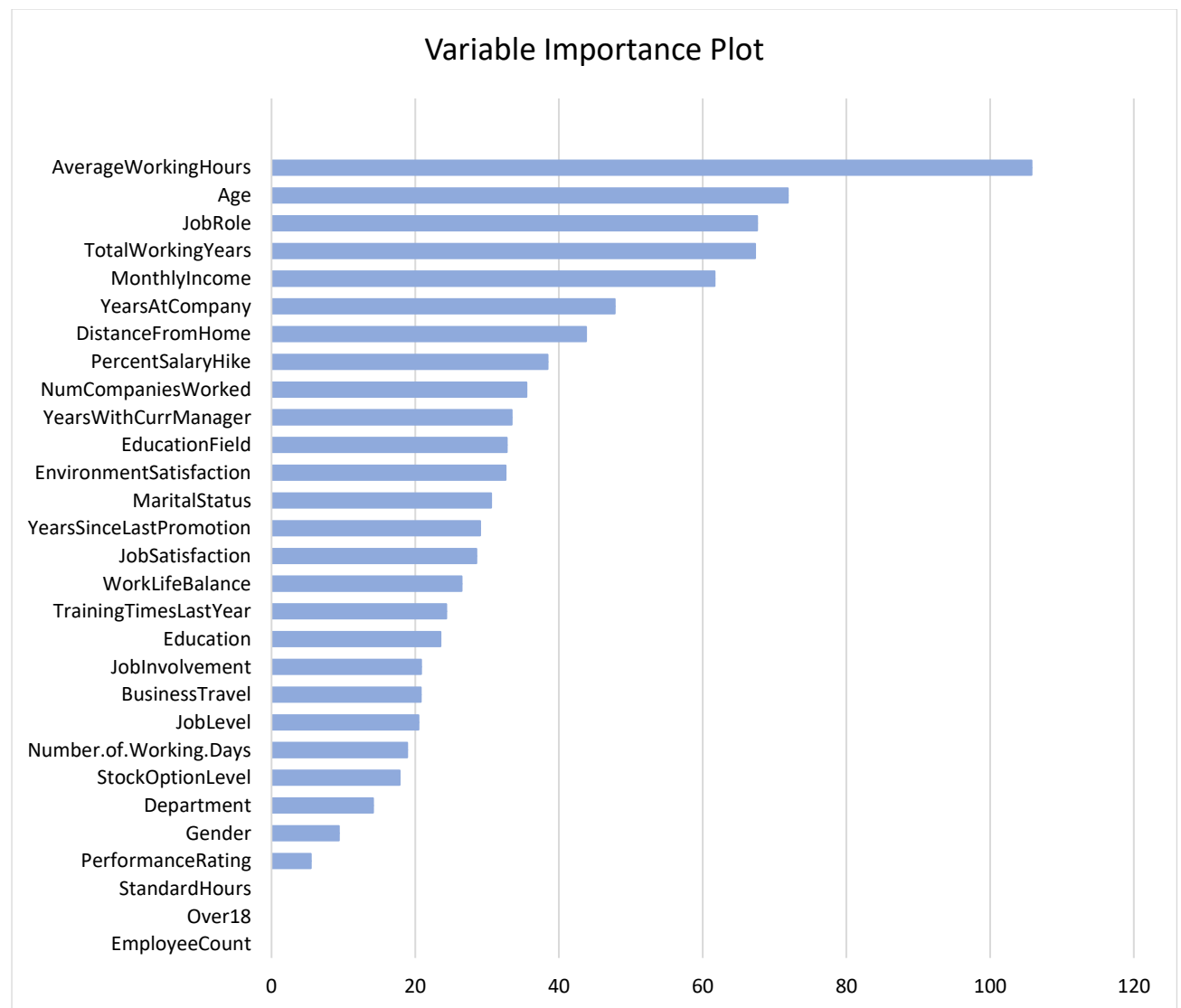


Figure 5

With maximum mean decrease in Gini index, the variable 'Average Working Hours' is the most important variable in identifying whether the person has attrition indicator as Yes or No. It is followed by the variables 'Age', 'Job Role', 'Total Working Years' and 'Monthly Income'.

The 10-fold CV for Random forest model fit on the data is 0.00162. (Accuracy: 0.998). This shows that the classifier works extremely well on the test data.

Though the prediction accuracy is very high Random Forest does not help us in interpreting the important factors. A tree fit on the data gives some idea about the splits made in the data. We assume that as Random Forest is a collection of many trees, this tree reflects some part of the relation in the Random Forest.

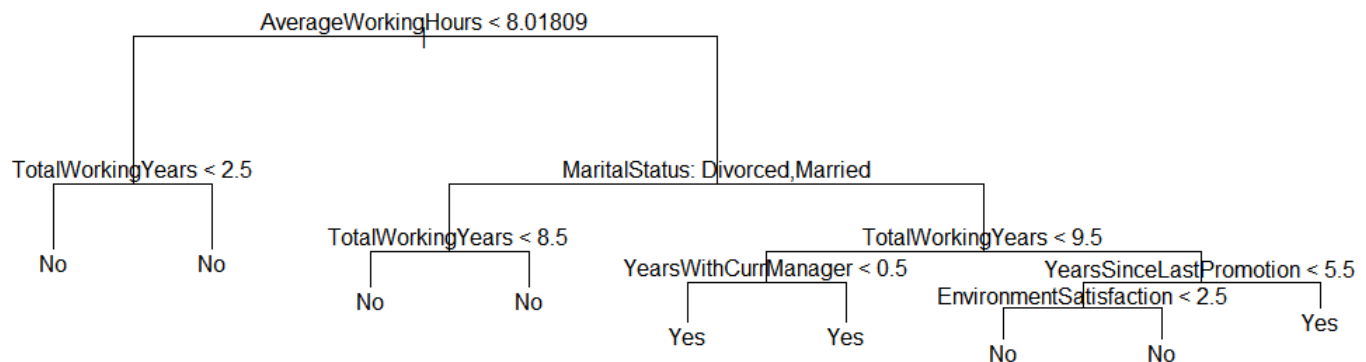


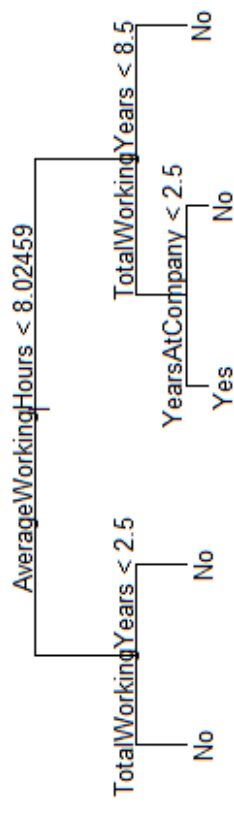
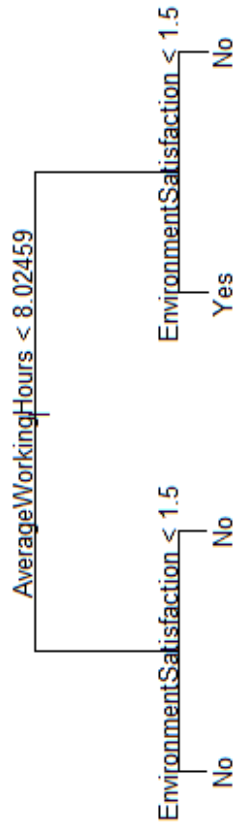
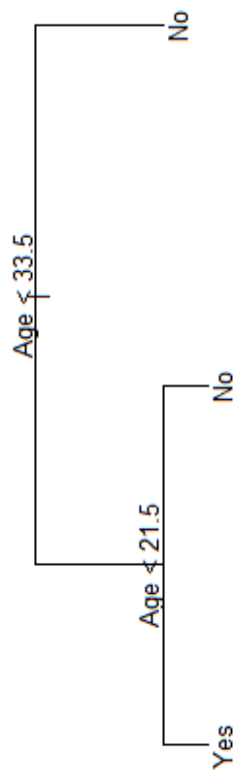
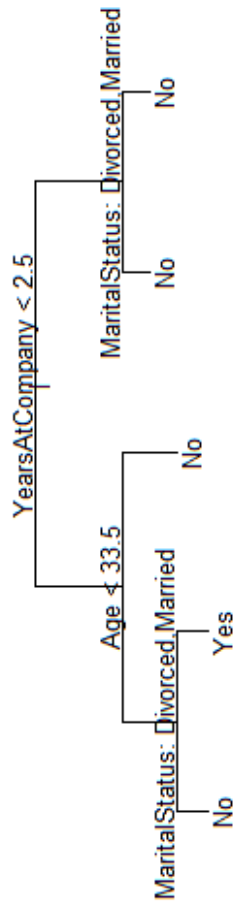
Figure 6

The single employees that work for more than 8 hours average daily, with total working years either less than 9.5 years, or more than 9.5 years but with more than 5.5 years since last promotion are at risk of leaving the company.

The above tree considers all the variables involved, as opposed to Random Forest, which considers only p (Here p is the parameter $mtry$ which is chosen to be 6 by tuning) and 100 such trees (parameter $ntrees = 100$). Thus we constructed and observed 100 trees with only 6 variables chosen randomly. These trees will mimic the Random Forest algorithm and give us some interpretable results. Most of the trees hence obtained are not favourable for interpretation as the tree fails to classify into two distinct classes i.e. only “No” predicted.

Some of the trees which were observed are plotted below.

The most important variables in the Variable importance plot (Fig 5) which are Average Working Hours and Age, are observed to be the primary split.



On the basis of the above trees, those at risk of leaving the company are :

i) Employees younger than 21.5 years

ii) Employees under the age 33.5 who are single and have joined the company less than 2.5 years ago

Both the cases suggest employees that are young are at risk. Younger employees may have different career goals such as experimenting with jobs before settling on one. They might leave the job for higher education or personal matters. A young employee may lack experience which may reflect in his performance.

iii) Employees working more than 8 hours average daily with low Environmental satisfaction

iii) Employees working more than 8 hours average daily and have joined the company less than 2.5 years ago, along with total working experience less than 8 years

This information will benefit the company in designing policies and rules limiting the daily working hours which will minimise hazards of overworking like burning out at a young age. Absence of promotion even though employee has been working in the company for almost a decade may be due unsatisfactory performance by the employee or the skills are being overlooked by the authority. The company can revise and improve their appraisal process.

The management can focus on the younger employees and their reasons for leaving the company.

CONCLUDING REMARKS

A major problem in high employee attrition is its cost to an organization. Job postings, hiring processes and training are some of the common expenses of losing employees and replacing them. Additionally, regular employee attrition prohibits the organization from increasing its collective knowledge base and experience. From this analysis, we have tried to identify the reasons for attrition, which will help the organization to build a healthy environment at the workplace, in order to get most of the their employees to stay.

We are proposing five point agenda for the organization to reduce the employee attrition –

- 1) The most important feature affecting the attrition is ‘Average Working Hours’. The employees who work for more than 8 hours a day are most likely to leave the job. Hence, the organization need to manage the workload of their employees in such a way that it should not require overtime. This will not only curb the attrition rate but also increase the productivity of an individual, which will cause acceleration in the growth of the organization.
- 2) The organization need to concentrate more on their younger employees. Many of the organizations are exploring the option of providing constantly changing roles, involving them in decision-making and by giving earlier jump in salary. These things need to done by the organization, otherwise it will have a hard time holding on to huge talent possessed by the young employees.
- 3) Another important feature contributing in attrition is ‘Total Working Years’. The employees who are working for many years in the same organization and getting frequent promotions are generally loyal with the organization and are less likely to leave the job or get fired. Hence, the organization need to motivate the inexperienced employees and give them chance to learn from their mistakes, which will develop the relationship of mutual trust between the organization and the employees.
- 4) Interestingly, Monthly income is not as important as we are expected. This means that employees are giving more value to workplace environment and job role than the salary. Hence, the organization may not need to change their salary structure if the overall environment is improved.
- 5) Attrition rate is much higher in the HR department than the Sales and R & D. Thus, the organization need to have some well-formed strategy for the HR employees, which will serve as a catalyst for reducing the attrition rate.